

PROJECT REPORT

MULTI-REGION DISASTER RECOVERY ARCHITECTURE FOR PAYMENT SYSTEMS (Project 1B)

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Executive Summary

PaySecure Gateway Private Limited is a fictional digital payment company that processes millions of financial transactions every day across India. The company currently operates its entire payment platform within a single AWS Region, creating a significant business continuity risk in the event of a regional outage. Any disruption could result in transaction failures, financial losses, regulatory penalties, and loss of customer trust.

This project proposes a production-grade Multi-Region Disaster Recovery (DR) Architecture designed to ensure high availability, business continuity, and regulatory compliance. The proposed solution uses AWS services such as Amazon Route 53, Amazon Aurora PostgreSQL, DynamoDB Global Tables, Amazon ElastiCache Redis, and Apache Kafka with cross-region replication and automated failover mechanisms. The architecture aims to achieve a Recovery Point Objective (RPO) of less than one minute and a Recovery Time Objective (RTO) of less than five minutes while maintaining uninterrupted payment processing services.

Introduction

The rapid growth of digital payments has increased the need for highly available and resilient cloud infrastructure. Financial institutions are expected to provide uninterrupted services even during infrastructure failures, natural disasters, or regional outages. Any downtime in payment systems can lead to financial losses, customer dissatisfaction, and regulatory non-compliance.

PaySecure Gateway Private Limited is a fictional cloud-native payment company that currently operates its services from a single AWS Region. This architecture introduces a single point of failure, making the platform vulnerable to regional disruptions. To overcome this challenge, a robust Multi-Region Disaster Recovery Architecture is required.

The proposed solution leverages AWS cloud services to replicate applications and data across multiple regions, enabling automatic failover and rapid recovery. The architecture is designed to achieve high availability, business continuity, and compliance with RBI guidelines, PCI-DSS standards, and data localisation requirements while ensuring minimal service interruption during disaster events.

Existing System and Challenges

PaySecure Gateway currently operates its payment processing platform entirely within a single AWS Region (Mumbai). While this architecture supports daily business operations, it creates a significant single point of failure. In the event of a regional outage caused by infrastructure failures, network disruptions, or natural disasters, the entire payment platform may become unavailable, leading to service interruptions and business losses.

The existing infrastructure lacks an automated disaster recovery mechanism, cross-region replication, and intelligent traffic failover capabilities. Database recovery depends on manual intervention, increasing recovery time and the risk of data loss. The absence of a secondary disaster recovery region also limits business continuity and regulatory preparedness.

In addition, the current architecture does not fully satisfy the disaster recovery expectations of financial institutions operating under RBI guidelines and PCI-DSS standards. Therefore, implementing a resilient Multi-Region Disaster Recovery Architecture has become essential to ensure uninterrupted payment services, improved availability, and enhanced customer trust.

Business Requirements

The primary objective of this project is to design a highly available and resilient Multi-Region Disaster Recovery Architecture for PaySecure Gateway to ensure uninterrupted payment processing services. The proposed solution must minimize downtime and data loss while maintaining regulatory compliance and operational efficiency.

The architecture should support automatic failover between AWS Regions, continuous data replication, and rapid service recovery during regional failures. It must achieve a Recovery Point Objective (RPO) of less than one minute and a Recovery Time Objective (RTO) of less than five minutes. The solution should also comply with RBI guidelines, PCI-DSS security standards, and data localisation requirements while ensuring secure handling of sensitive payment information.

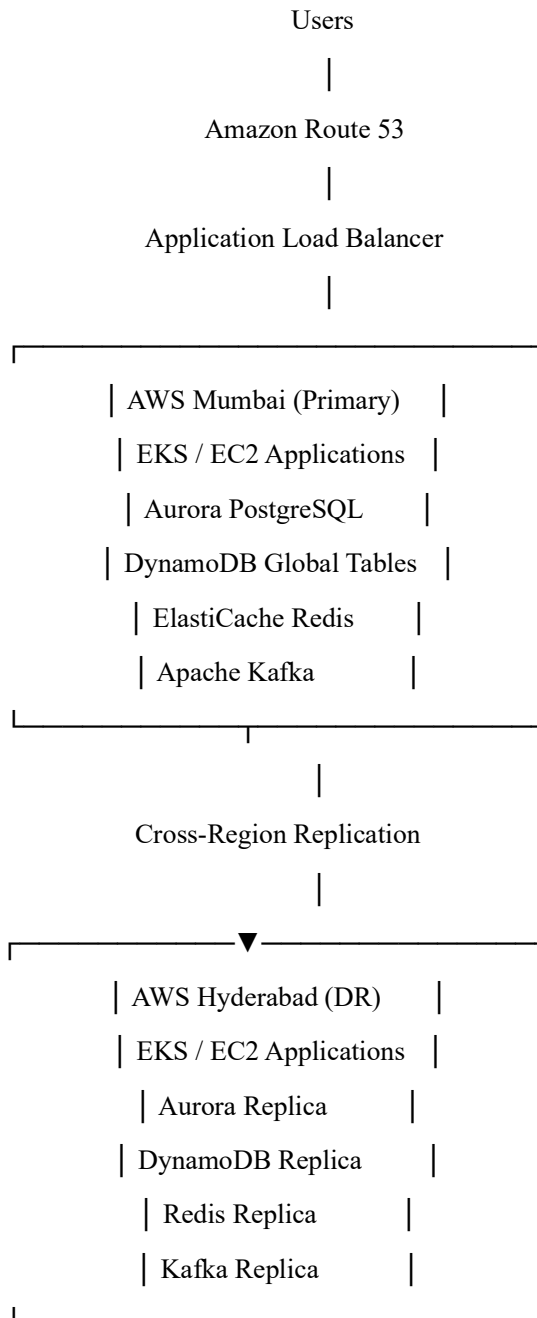
Furthermore, the disaster recovery framework should improve business continuity, increase customer trust, and provide a scalable infrastructure capable of handling future growth in transaction volumes without affecting system performance.

Proposed Multi-Region Disaster Recovery Architecture

The proposed solution introduces a production-grade Multi-Region Disaster Recovery (DR) Architecture that ensures continuous payment processing even during a regional outage. The architecture uses **AWS Mumbai Region as the Primary Region** and **AWS Hyderabad Region as the Disaster Recovery Region**. All critical application components and databases are replicated across both regions to maintain high availability and business continuity.

The primary region handles all live payment transactions under normal operating conditions, while the disaster recovery region remains synchronized through continuous cross-region replication. In the event of a failure in the primary region, Amazon Route 53 automatically redirects user traffic to the disaster recovery region, ensuring minimal downtime and uninterrupted services.

The architecture includes Amazon Aurora PostgreSQL for transactional data, DynamoDB Global Tables for globally replicated metadata, Amazon ElastiCache Redis for caching, Apache Kafka for event streaming, and Amazon S3 Cross-Region Replication for backup storage. This design enables the platform to achieve an RPO of less than one minute and an RTO of less than five minutes while maintaining compliance with RBI and PCI-DSS requirements



Active-Passive Deployment Strategy

The proposed disaster recovery solution follows an **Active-Passive deployment strategy** to ensure high availability and business continuity. In this architecture, the **AWS Mumbai Region** serves as the primary region where all payment transactions and application services are processed under normal conditions. The **AWS Hyderabad Region** acts as the disaster recovery region and remains on standby while continuously receiving replicated data from the primary region.

If the primary region experiences a failure due to infrastructure issues, network outages, or natural disasters, **Amazon Route 53** automatically redirects user traffic to the disaster recovery region. Since the application data and databases are already synchronized, services can be restored quickly with minimal downtime and data loss.

This deployment strategy helps achieve an **RPO of less than one minute** and an **RTO of less than five minutes** while ensuring uninterrupted payment processing. It also supports **RBI guidelines, PCI-DSS compliance, and business continuity requirements**, making it a reliable and cost-effective disaster recovery solution for PaySecure Gateway.

Active-Active Deployment Strategy

The **Active-Active deployment strategy** allows both the **AWS Mumbai Region** and **AWS Hyderabad Region** to remain active simultaneously, sharing application traffic and processing payment transactions together. This approach provides maximum availability and improves system resilience by distributing workloads across multiple regions.

If one region becomes unavailable due to a failure or disaster, the other region continues processing transactions without interrupting customer services. Continuous data replication and synchronization between both regions ensure consistency and reduce the risk of data loss. Load balancing and intelligent traffic routing help maintain optimal performance during normal operations as well as during failover events.

Although the Active-Active model offers higher availability and faster recovery than the Active-Passive approach, it is more complex to manage and requires additional infrastructure resources. However, it is highly suitable for mission-critical payment systems where continuous service availability is essential.

Database Replication Strategy

The proposed disaster recovery architecture uses **cross-region database replication** to ensure data availability and minimize the risk of data loss during a regional outage. The primary database is hosted in the **AWS Mumbai Region**, while a continuously synchronized replica is maintained in the **AWS Hyderabad Region**. This replication process keeps both regions updated with the latest transaction data.

Amazon Aurora PostgreSQL is used for storing payment transaction records and customer information. **DynamoDB Global Tables** replicate metadata and application state across regions, enabling fast access and high availability. **Amazon ElastiCache Redis** synchronizes frequently accessed cache data, while **Apache Kafka** replicates event streams and payment messages between regions to maintain transaction consistency.

With continuous replication and automatic synchronization, the architecture supports a **Recovery Point Objective (RPO) of less than one minute**, ensuring that almost no transaction data is lost during a disaster. This strategy provides reliable data protection, faster recovery, and uninterrupted payment processing services.

DNS Failover and Traffic Management

The proposed disaster recovery architecture uses **Amazon Route 53** to provide intelligent DNS failover and efficient traffic management across multiple AWS Regions. Under normal operating conditions, all user requests and payment transactions are directed to the **AWS Mumbai Region**, which acts as the primary production environment.

Amazon Route 53 continuously performs health checks on the primary infrastructure. If the primary region becomes unavailable due to a regional outage, application failure, or network disruption, Route 53 automatically detects the issue and redirects incoming traffic to the **AWS Hyderabad Disaster Recovery Region**. This automated failover process minimizes service interruption and ensures that customers continue to access payment services without significant downtime.

The traffic management mechanism improves application availability, supports rapid disaster recovery, and helps achieve the target **Recovery Time Objective (RTO) of less than five minutes**. By eliminating manual intervention during failover events, the solution enhances business continuity, operational reliability,

and customer experience while supporting regulatory compliance for critical financial services.

Security and Compliance

Security and regulatory compliance are essential components of the proposed Multi-Region Disaster Recovery Architecture. The solution is designed to protect sensitive payment information while ensuring that all services operate according to industry regulations and security standards. Strong access control, encryption, continuous monitoring, and secure data replication help maintain the confidentiality and integrity of financial data.

All payment data is encrypted both **at rest and in transit** using AWS security services and industry-standard encryption techniques. Identity and Access Management (IAM) policies enforce least-privilege access, while AWS CloudTrail and CloudWatch provide continuous logging and monitoring of system activities. Regular security audits and automated monitoring help identify potential threats and maintain operational security.

The proposed architecture also supports **RBI guidelines, PCI-DSS requirements, and data localisation regulations** by ensuring secure storage of payment information within approved regions and implementing appropriate disaster recovery controls. These security measures improve customer trust, reduce operational risk, and strengthen the overall resilience of the payment platform.

Disaster Recovery Runbooks

A Disaster Recovery Runbook is a documented set of procedures that guides the operations team during a disaster or regional outage. It provides a structured recovery process to restore critical payment services quickly while minimizing downtime and data loss. Having a predefined runbook ensures that recovery activities are performed consistently and efficiently during emergency situations.

When a failure occurs in the primary AWS Mumbai Region, the operations team first verifies the incident using monitoring tools such as **Amazon CloudWatch** and **AWS Health Dashboard**. After confirming the outage, **Amazon Route 53** automatically redirects user traffic to the AWS Hyderabad Disaster Recovery Region. The application services, databases, and supporting components in the

DR region become active and continue processing transactions without major interruption.

After the primary region is restored, data consistency is verified, application health checks are completed, and traffic is gradually shifted back to the primary environment through a controlled failback process. Regular disaster recovery drills and testing ensure that the runbooks remain effective and that the organization is prepared for real-world disaster scenarios.

Cost Analysis and Business Benefits

Implementing a Multi-Region Disaster Recovery Architecture requires additional cloud infrastructure and replication services, resulting in an increase in operational costs. However, the investment provides significant long-term benefits by reducing the financial impact of unexpected outages, preventing transaction failures, and ensuring uninterrupted payment services. The cost of maintaining a disaster recovery environment is considerably lower than the potential losses caused by prolonged downtime.

The proposed architecture improves overall business continuity by enabling automatic failover, continuous data replication, and rapid service recovery. It helps the organization maintain customer confidence, protect its reputation, and meet regulatory expectations during disaster situations. The architecture also reduces operational risks by minimizing manual recovery activities and improving infrastructure reliability.

From a business perspective, the solution supports future scalability, enhances service availability, and strengthens the organization's disaster preparedness. By achieving high availability and faster recovery, PaySecure Gateway can continue processing payment transactions efficiently even during regional failures, ensuring a reliable experience for customers and merchants.

Conclusion

The proposed **Multi-Region Disaster Recovery Architecture** provides a reliable and resilient solution for ensuring uninterrupted payment processing services during regional failures or unexpected disasters. By utilizing AWS cloud technologies such as **Amazon Route 53**, **Amazon Aurora PostgreSQL**, **DynamoDB Global Tables**, **Amazon ElastiCache Redis**, and **Apache Kafka**,

the architecture enables continuous data replication, automatic failover, and rapid service recovery.

The solution is designed to achieve a **Recovery Point Objective (RPO) of less than one minute** and a **Recovery Time Objective (RTO) of less than five minutes**, ensuring minimal data loss and service disruption. It also strengthens business continuity, improves customer trust, and supports compliance with **RBI guidelines, PCI-DSS standards, and data localisation requirements**.

Overall, the proposed disaster recovery strategy offers a scalable, secure, and cost-effective approach for protecting critical payment infrastructure and maintaining high availability in modern cloud environments. It enables PaySecure Gateway to deliver reliable financial services while remaining prepared for future operational challenges and disaster scenarios.

Future Enhancements

The proposed Multi-Region Disaster Recovery Architecture can be further enhanced by adopting advanced cloud-native technologies and intelligent automation mechanisms. Future improvements may include the implementation of AI-driven monitoring systems for predictive failure detection, automated disaster recovery testing, and self-healing infrastructure using Infrastructure as Code (IaC) tools such as Terraform and AWS CloudFormation.

The organization can also consider implementing a fully **Active-Active multi-region architecture** to distribute workloads across regions and further reduce recovery time. Advanced security services such as AWS GuardDuty, AWS Security Hub, and AWS Shield Advanced can be integrated to strengthen cyber resilience against sophisticated attacks.

Regular disaster recovery drills, continuous compliance monitoring, and automated infrastructure validation can further improve operational readiness. These enhancements will help the organization achieve higher availability, improved scalability, and stronger business continuity while supporting future growth in digital payment transactions.

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